

1. Dulmarka Guud ee Nidaamka (System Overview)

DUGSI PRO 2026 waa nidaam dhammeystiran oo casri ah oo loogu talagalay maamulka dugsiyada, machadyada, iyo xarumaha waxbarashada (All-in-One Cloud School ERP). Nidaamku wuxuu si hufan u xalliyaa maamulka ardayda, xaadirinta labada xilli, biilasha & rasiidhada, fasallada, maaddooyinka, imtixaanaadka, iyo warbixinada PDF & WhatsApp.

Qaab-dhismeedka Tiknoolajiyadda (Tech Stack)

Qaybta (Layer)	Tiknoolajiyadda (Technology)	Ujeeddada & Doorka (Role)
Frontend UI	React 19, TypeScript, Vite 6, Tailwind CSS v4	Shaashadaha casriga ah, xawaare sare, mobile responsive
Animations & Icons	Motion (Framer Motion), Lucide React	Dhaqdhaqaaqa quruxda badan iyo calaamadaha casriga ah
Charts & Data	Recharts Interactive Analytics	Jaantusyada maaliyadda, xaadirinta, iyo natiijooyinka ardayda
Backend Engine	Node.js & Express 4 (TypeScript)	RESTful API routes, xaqiijinta amniga iyo xogta dugsiga
Cloud Database	Supabase (PostgreSQL Cloud)	Kaydka rasmiga ah ee daruuraha oo leh Multi-Tenant isolation
Local Fallback DB	Local JSON DB (database.json)	Kayd offline degdeg ah haddii internetku go'o oo auto-sync leh
PDF & Reports	jsPDF, jsPDF-AutoTable, SheetJS (XLSX)	Daabacaadda kaararka natiijada, rasiidhada, iyo Excel import/export
AI & Caqli Gacood	@google/genai (Google Gemini AI)	Falanqaynta xogta dugsiga iyo caqliga macmalka ah

Amniga iyo Kala-soocnaanta Dugsiyada (Multi-Tenant Isolation)

Dugsiyadu waxay leeyihiin xog gebi ahaanba kala xiran oo madax-bannaan. Codsii kasta oo frontend-ka ka yimaada wuxuu wataa cinwaanka email-ka dugsiga (X-School-Email). Server-ku wuxuu u beddelaa aqoonsi gaar ah (school_id), taas oo xaqiijinaysa in dugsi kale uusan marnaba arki karin ama taaban karin ardayda, lacagaha, ama natiijooyinka dugsi kale.

Module-ka	Sharaxaadda Shaqada iyo Astaamaha (Description & Capabilities)
1. Dashboard	Guddi kormeer oo muujinaya tirada guud ee ardayda, boqolleyda xaadirinta maanta, dakhliga bishan, iyo digniinaha lacagaha dhiman.
2. Maamulka Ardayda	Diiwaangelinta ardayda, sawirka, taleefanka waalidka, fasalka, ka-hortagga duplicate-ka, iyo soo gelinta/dejinta Excel (.xlsx) & PDF.
3. Xaadirinta Maalinlaha	Xaadirinta labada xilli (Nasashada ka hor & Nasashada ka dib), xaaladaha: Jooga, Maqan, Soo daahay, Fasax. Farriin WhatsApp degdeg ah.
4. Lacagaha & Biilasha	Abuurista biilasha ardayda/fasalka (Bulk Invoicing), bixinta buuxda/qaybta ah, taariikhda bixinta, iyo Rasiidka Rasmiga ah ee PDF.
5. Fasallada & Qolalka	Maamulka fasallada, macallinka masuulka ah, lambarka qolka, iyo xisaabinta tooska ah ee tirada ardayda fasal kasta ku jirta.
6. Maaddooyinka	Diiwaangelinta maaddooyinka, koodhadhka maaddooyinka, macallimiinta dhiga, iyo fasallada loo xilsaaray.
7. Imtixaanaadka	Gelinta natiijooyinka imtixaanaadka (Midterm, Final, Monthly Test), xisaabinta tooska ah ee darajooyinka A, B, C, D, iyo Fail.
8. Warbixinada & Kaarka	Warbixinta waxqabadka fasalka (Class Performance) iyo Kaarka Natiijada Ardayga (Student Report Card) oo PDF ah oo waalidka loo diri karo.
9. Qaabeynta (Settings)	Beddelka magaca dugsiga, lacagta (Currency), qiimaha caadiga ah ee fiiga, xadka buundooyinka, iyo Light/Dark Mode.
10. Backup & Reset	Soo dejinta/gelinta xogta dugsiga oo JSON ah, xiriirka tooska ah ee Supabase, iyo tirtiridda xogta dugsiga oo kaliya haddii loo baahdo.

Miisaska Kaydka Xogta ee Supabase (PostgreSQL Schema)

Miiska (Table Name)	Furaha Sare (PK)	Columns-ka Muhiimka ah	Sharaxaadda
dugsiga_users	email	password, verified, created_at	Akoonnada maamulayaasha dugsiyada
dugsiga_students	id	school_id, full_name, class, gender, phone, photo	Diiwaanka dhammaan ardayda dugsiga
dugsiga_classes	id	school_id, class_name, teacher_name, room_number	Liiska fasallada dugsiga
dugsiga_subjects	id	school_id, subject_name, subject_code, class_name	Maaddooyinka waxbarashada
dugsiga_exam_scores	id	school_id, student_id, class_name, marks, grade	Natiijooyinka imtixaanaadka ardayda
dugsiga_attendance	Composite	status, timestamp, session_type	Diiwaanka xaadirinta maalinlaha ah
dugsiga_fees	id	school_id, student_id, month, amount, paid_amount, history	Biilasha iyo xisaabaadka lacag-bixinta
dugsiga_settings	Composite	value (JSONB)	Qaabeynta dugsiga (Magaca, Lacagta, iwm)

Liiska REST API Endpoints-ka Server-ka

Method	Endpoint Path	Headers / Query	Ujeeddada (Function)
POST	/api/auth/signup	Body: email, password	Diiwaangelinta dugsi cusub
POST	/api/auth/login	Body: email, password	Galitaanka nidaamka
GET / POST	/api/students	Header: X-School-Email	Soo saarista & ku darista ardayda
PUT / DELETE	/api/students/:id	Header: X-School-Email	Wax ka beddelka & tirtiridda ardayga
GET / POST	/api/attendance	Query: date, session_type	Soo qaadista & kaydinta xaadirinta
GET / POST	/api/fees	Header: X-School-Email	Soo saarista & abuurista biilasha
PUT / DELETE	/api/fees/:id	Header: X-School-Email	Bixinta lacagta & soo saarista rasiidka
GET / POST	/api/classes	Header: X-School-Email	Maamulka fasallada dugsiga
GET / POST	/api/subjects	Header: X-School-Email	Maamulka maaddooyinka waxbarashada
GET / POST	/api/exams	Header: X-School-Email	Maamulka natiijooyinka imtixaanaadka
GET / PUT	/api/settings	Header: X-School-Email	Soo qaadista & keydinta qaabeynta dugsiga
GET	/api/db/status	None	Hubinta xiriirka Supabase Cloud Database
POST	/api/reset	Header: X-School-Email	Tirtiridda xogta dugsiga gelaya (Factory Reset)

Furayaasha Sirta ah & Environment Variables (.env)

Variable Name	Qiimaha / Tusaalaha (Value / Example)	Ujeeddada iyo Amniga (Security Note)
GEMINI_API_KEY	AlzaSy... (Laga helo Google AI Studio)	Furaha caqliga gacoodka Google Gemini (Server-side secret)
APP_URL	https://ais-dev-...run.app	Cinwaanka rasmiga ah ee nidaamku ka shaqeynayo
SUPABASE_URL	https://mdvfcqujqjnvfpzowayo.supabase.co	Cinwaanka mashruuca Cloud Database ee Supabase
SUPABASE_ANON_KEY	eyJhbGciOi... (Anon JWT Token)	Furaha caadiga ah ee la wadaagi karo Supabase SDK
SUPABASE_SECRET_KEY	eyJhbGciOi... (Service Role Key)	Furaha qarsoon ee maamulaha sarre ee Supabase (Super Admin)
SUPABASE_PUBLISHABLE_KEY	sb_publish_... / anon key	Furaha publishable-ka ah ee ku habboon client SDKs
SUPABASE_JWKS_URL	Supabase Auth JWKS Endpoint	URL-ka xaqijinta tokens-ka JWT
SMTP_HOST	smtp.gmail.com	Server-ka loo isticmaalo dirista email-lada
SMTP_PORT	587 (TLS) ama 465 (SSL)	Port-ka ammaan ah ee SMTP email
SMTP_USER	your_email@gmail.com	Cinwaanka email-ka diraha rasmiga ah
SMTP_PASS	your_google_app_password	Furaha sirta ah ee Google App Password ee email-ka
SMTP_FROM	your_email@gmail.com	Cinwaanka ka muuqanaya sanduuqa email-ka qaataha

Tallaabooyinka Ku-rakibidda iyo Kicinta (Installation & Running)

- ```
1. Ku shub dhammaan xirmooyinka lagama maarmaanka ah
npm install

2. Kici server-ka horumarinta (Development Mode)
npm run dev

3. U diyaari una kici shaashadda rasmiga ah (Production Deployment)
npm run build && npm start
```